Yusen Wu

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Education

University of Virginia (UVA), Charlottesville, VA, USA 2025.08 – Present
Ph.D. in Computer Science Advisor: Yue Cheng

Sichuan University (SCU), Chengdu, China 2021.09 – 2025.06
B.E. in Computer Science and Technology (Top-notch Program)

Publications

- **[SIGMOD'27]** Tingfeng Lan, **Yusen Wu**, Bin Ma, Zhaoyuan Su, Rui Yang, Tekin Bicer, Masahiro Tanaka, Olatunji Ruwase, Dong Li, Yue Cheng. *ZenFlow: Enabling Stall-Free Offloading for LLM Training.* ACM SIGMOD International Conference on Management of Data (SIGMOD), 2027. To appear.
- **[Preprint] Yusen Wu**, Tingfeng Lan, Benjamin Carver, Haoliang Wang, Geoffrey Fox, Yue Cheng. *LooseLeaf: Enabling Serverless GPUs for Computational Notebooks.* Under revision at VLDB 2027.

Research Experience

LooseLeaf: Enabling Serverless GPUs for Computational Notebooks 2025.09 – 2026.09
Project Lead Advisor: Yue Cheng (UVA)

- Built LooseLeaf, a serverless GPU platform that allocates GPUs per notebook cell, acquiring a GPU only when a GPU cell runs and reclaiming it right after, instead of pinning one to the whole session.
- Characterized ∼5,000 real-world notebooks and showed that a GPU cell needs only a small part of the session state, which LooseLeaf derives by static analysis and loads into a GPU worker while the user edits.
- Designed a per-variable checkpoint store that preserves object sharing across variables, and a CUDA-safe import cache that forks GPU workers with their packages preloaded.
- Started 95–98% of GPU cells within one second and cut GPU-hours by 43–70% versus session-pinned GPUs and by 32–55% versus NotebookOS (ASPLOS'26).

ZenFlow: Enabling Stall-Free Offloading for LLM Training 2024.10 – 2025.08
Lead Implementer Advisors: Yue Cheng (UVA) and Dong Li (UC Merced)

- Co-developed ZenFlow, an offloading framework for LLM fine-tuning that updates important gradients on the GPU right away and applies the rest asynchronously on the CPU, reducing GPU stalls.
- Led the ∼11K-line implementation on DeepSpeed, including gradient selection across sharded GPUs, and proposed a column-major gradient layout that makes channel-wise selection efficient.
- Achieved up to 5× speedup, 85% fewer GPU stalls, and 72% lower cloud GPU cost with no accuracy loss.

Open Source & Deployment

ZenFlow in DeepSpeed (#7391, #7516), featured on the PyTorch Blog 2025.08 – 2025.10
LooseLeaf at looseleafcloud.org, open to the UVA community 2026.10 – Present

Honors and Awards

National Scholarship, Ministry of Education of China (top 0.2% nationwide) 2021 – 2022
First Prize Scholarship, Sichuan University 2023 – 2024
Merit Student, Sichuan University (three times) 2021 – 2024

Skills

- **Programming:** C/C++, CUDA, Python
- **Systems and Tools:** PyTorch, DeepSpeed, Ray, Redis, Slurm